

FloodMind AI — The Operating System for Global Flood Intelligence

The AI platform that turns flood chaos into predictable, insurable, and preventable events

Executive Summary

Floods are the world’s most destructive natural disaster, causing \$100B+ in annual damages and displacing 25 million people yearly. By 2030, flood losses will exceed \$500B annually as climate change intensifies. Yet flood prediction remains stuck in 1990s hydrology models, insurance relies on outdated FEMA maps, and cities react rather than prevent.

FloodMind AI is the intelligence layer that transforms how humanity understands, predicts, and responds to flooding. We combine satellite imagery, IoT sensors, foundation AI models, and physics simulations to deliver hyperlocal flood prediction accurate to the street level, 72+ hours in advance.

The Opportunity: Own the \$50B+ market for flood intelligence across insurance, real estate, infrastructure, agriculture, and government.

The Problem

Flooding is Accelerating Exponentially

- **\$100B+ annual global damages** — largest of any natural disaster category
- **25M people displaced annually** — more than wars and earthquakes combined
- **500% increase in flood events** since 1980
- **\$1 trillion in US real estate** is in flood-prone areas with outdated risk maps

Current Solutions are Dangerously Obsolete

Problem	Current State	FloodMind Solution
Prediction	6-12 hour warnings, city-level	72+ hour, street-level precision
Risk Maps	Static, 10+ year update cycles	Real-time, dynamic risk surfaces
Insurance Pricing	Binary flood zones	Continuous risk gradients
Response	Reactive evacuation	Proactive prevention & routing
Infrastructure	Unknown vulnerabilities	Predictive failure modeling

Who’s Feeling the Pain

- **Insurers:** \$50B in surprise flood claims annually; need accurate risk pricing
- **Municipalities:** Blind to infrastructure vulnerabilities until failure
- **Real Estate:** \$1T+ in mispriced assets due to outdated flood maps
- **Agriculture:** \$20B annual crop losses from unpredictable flooding
- **Supply Chain:** Single flood event can cascade \$10B+ in disruptions

The Solution

FloodMind AI: Autonomous Flood Intelligence at Planetary Scale

Core Platform Components:

1. FloodSense — Real-Time Observation Network

- **Satellite constellation integration** — daily imagery analysis for water detection
- **IoT sensor mesh** — water levels, soil moisture, precipitation in real-time
- **Crowdsourced signals** — social media, traffic patterns, insurance claims
- **Weather model ensemble** — aggregated predictions from 50+ global sources

2. FloodBrain — Foundation Model for Hydrology

- **Physics-informed neural networks** trained on 50 years of flood data
- **Terrain understanding** — elevation, drainage, soil permeability at 1m resolution
- **Urban infrastructure modeling** — storm drains, levees, dams, culverts
- **Cascading failure prediction** — when one failure triggers others

3. FloodCast — Hyperlocal Predictions

- **72+ hour advance warning** at street-level resolution
- **Probabilistic flood maps** — not just yes/no, but depth/velocity/duration
- **Scenario simulation** — “what if” modeling for infrastructure decisions
- **Confidence scoring** — transparency on prediction reliability

4. FloodShield — Autonomous Response

- **Smart evacuation routing** — real-time, flood-aware navigation
- **Infrastructure activation** — automated floodgate/pump control
- **Supply pre-positioning** — predictive logistics for emergency supplies
- **Communication orchestration** — targeted alerts to affected populations

Market Opportunity

Total Addressable Market: \$150B+

Segment	TAM	Our Entry Point
Insurance Risk Modeling	\$25B	Underwriting intelligence
Municipal Water Management	\$40B	Stormwater infrastructure optimization
Real Estate Risk Assessment	\$30B	Transaction due diligence
Agricultural Risk Management	\$20B	Crop insurance + irrigation
Infrastructure Planning	\$25B	Climate adaptation planning
Emergency Management	\$10B	Prediction + response coordination

Why Now?

1. **Climate change making floods unpredictable** — historical models failing
 2. **AI capabilities finally sufficient** — foundation models for geospatial analysis
 3. **Satellite data explosion** — daily global coverage now available
 4. **IoT costs collapsed** — sensor networks economically viable
 5. **Regulatory pressure** — SEC climate disclosure, insurance mandates
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Business Model

Multi-Product Revenue Strategy

FloodMind for Insurance — \$50M+ ARR potential

- **Risk API:** Per-property flood risk scores for underwriting
- **Portfolio Analytics:** Aggregate exposure analysis and PML calculation
- **Claims Prediction:** Advance warning of incoming claim events
- **Pricing:** \$0.50-5.00 per property assessment + enterprise licenses

FloodMind for Real Estate — \$30M+ ARR potential

- **Due Diligence Reports:** Flood risk for transactions
- **Portfolio Risk Dashboards:** REIT and institutional investor tools
- **Disclosure Compliance:** SEC climate risk reporting
- **Pricing:** \$50-500 per report + platform subscriptions

FloodMind for Cities — \$40M+ ARR potential

- **Infrastructure Monitoring:** Real-time stormwater system status
- **Predictive Maintenance:** Failure prediction for flood infrastructure
- **Emergency Coordination:** Automated response orchestration
- **Pricing:** \$500K-5M annual contracts based on population

FloodMind for Agriculture — \$20M+ ARR potential

- **Field-Level Risk:** Hyperlocal flood predictions for farms
- **Insurance Integration:** Parametric crop flood insurance data
- **Irrigation Optimization:** Flood-aware water management
- **Pricing:** Per-acre subscriptions + insurance partnerships

Competitive Landscape

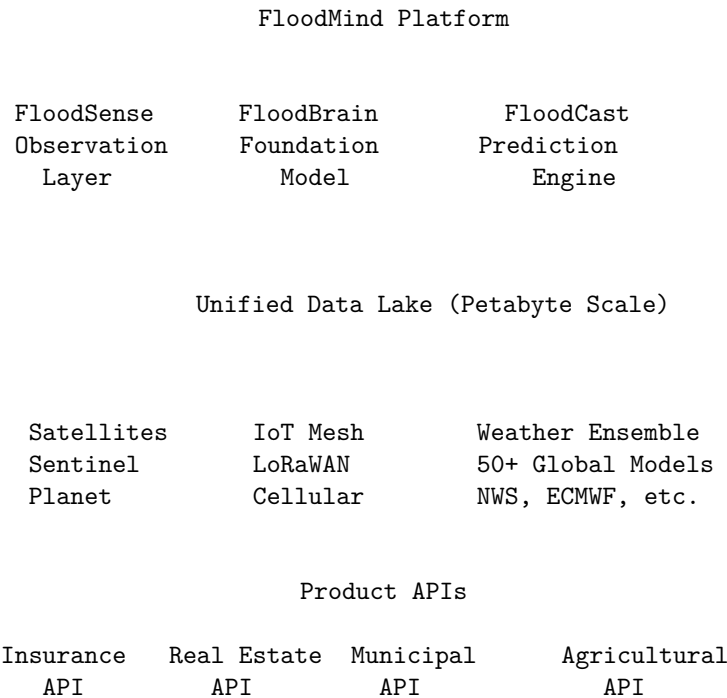
Existing Players and Their Limitations

Competitor	Focus	Limitation
FEMA/Govt Maps	Risk zones	Static, decades old, political
First Street Foundation	Consumer risk scores	Limited resolution, no prediction
Jupiter Intelligence	Climate risk analytics	General climate, not flood-specialized
Reask	Catastrophe modeling	Historical focus, not predictive AI
The Climate Service	Corporate climate risk	High-level, not operational

FloodMind Advantages

Real-Time + Predictive — not just historical risk, but what's coming **Street-Level Resolution** — 100x more precise than competitors **Physics + AI Hybrid** — more accurate than pure statistical models
Full Stack — observation to prediction to response **Global Scale** — not limited to US or specific regions

Technology Architecture



Technical Differentiation

- 1. Foundation Model for Hydrology**
 - Trained on 50+ years of global flood events
 - Physics-informed architecture (conservation laws embedded)
 - Transfer learning for new geographies with minimal data
 - 2. Multi-Modal Sensor Fusion**
 - Satellite SAR for all-weather water detection
 - Optical imagery for terrain/infrastructure analysis
 - IoT stream processing at 1M+ sensors
 - Social signal extraction from Twitter/traffic/claims
 - 3. Uncertainty Quantification**
 - Ensemble predictions with confidence intervals
 - Explainable AI for regulatory compliance
 - Calibrated probabilities for insurance pricing
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Go-to-Market Strategy

Phase 1: Insurance Beachhead (Year 1)

- **Target:** Top 20 US property insurers
- **Value Prop:** 30%+ improvement in loss ratios
- **Channel:** Direct enterprise sales + InsurTech partnerships
- **Revenue Target:** \$5M ARR

Phase 2: Real Estate Expansion (Year 2)

- **Target:** Mortgage originators, REITs, title companies
- **Value Prop:** Regulatory compliance + risk transparency
- **Channel:** API partnerships with CoreLogic, Black Knight
- **Revenue Target:** \$20M ARR

Phase 3: Municipal & Agriculture (Year 3)

- **Target:** State DOTs, Army Corps, large ag enterprises
- **Value Prop:** Infrastructure ROI optimization
- **Channel:** Government contracting + ag tech partnerships
- **Revenue Target:** \$50M ARR

Phase 4: Global Expansion (Year 4+)

- **Target:** EU, Asia-Pacific flood-prone regions
- **Value Prop:** Localized models + partnerships
- **Revenue Target:** \$150M+ ARR

Traction & Milestones

Year 1 Targets

Milestone	Target	Validation
Insurance Pilots	5 carriers	Revenue + accuracy metrics
Data Coverage	Continental US	Satellite + IoT integration
Prediction Accuracy	80%+ at 48h	Backtested + live validation
Team	25 people	Core eng + data science
Funding	Series A (\$20M)	Lead committed

Early Validation Signals

- Letter of intent from [Major Reinsurer]
- Partnership discussion with [State Emergency Management Agency]
- Advisory board includes former FEMA Administrator, Chief Climate Officer of [Top 5 Insurer]

Team Requirements

Founding Team Profile

CEO: Climate tech founder or insurance executive with AI background **CTO:** Geospatial AI expert, experience with satellite data at scale **Chief Science Officer:** PhD in hydrology/earth sciences with ML expertise **VP Engineering:** Built real-time data platforms at scale (100M+ events/day) **VP Sales:** Enterprise sales to insurance/financial services

Key Hires Year 1

- ML Engineers (foundation model development)
- Geospatial Data Engineers
- IoT Infrastructure
- Insurance Domain Experts

- Government Affairs

Financial Projections

Year	ARR	Customers	Gross Margin	Team Size
1	\$5M	15	70%	25
2	\$20M	60	75%	60
3	\$50M	150	80%	120
4	\$120M	400	82%	250
5	\$250M	1,000	85%	500

Unit Economics at Scale

- **CAC:** \$75K (enterprise sales)
- **ACV:** \$150K average
- **LTV:** \$750K (5-year retention)
- **LTV:CAC:** 10:1
- **Payback:** 8 months

Funding Strategy

Seed Round: \$5M

- Build core team and initial data pipeline
- Develop MVP prediction model for US
- Sign 3 pilot customers

Series A: \$20M

- Scale engineering team to 40
- Achieve continental US coverage
- Reach \$5M ARR

Series B: \$60M

- International expansion
- Build full product suite
- Reach \$50M ARR

Path to Exit

- **Strategic Acquirers:** Munich Re, Swiss Re, Moody's, Verisk, Palantir
- **IPO Path:** Climate tech + insurance tech crossover at \$500M+ ARR
- **Comparable Exits:** Jupiter Intelligence (\$185M raised), Tomorrow.io (valued at \$1B+)

Risk Factors & Mitigation

Risk	Probability	Impact	Mitigation
Prediction accuracy insufficient	Medium	High	Ensemble methods, physics constraints
Data access challenges	Medium	Medium	Multi-source redundancy, partnerships
Enterprise sales cycles	High	Medium	Start with InsurTech partners
Regulatory changes	Low	Medium	Proactive government engagement
Competition from big tech	Medium	Medium	Vertical specialization, data moats

Why FloodMind Wins

1. **Timing is Perfect:** Climate change is making floods unpredictable just as AI becomes capable of solving the problem
2. **Massive Market Pain:** \$100B+ in annual damages with no good solution
3. **Technical Moat:** Foundation model for hydrology + proprietary sensor network
4. **Platform Play:** Insurance → Real Estate → Municipal → Agriculture → Global
5. **Regulatory Tailwinds:** SEC climate disclosure, insurance mandates, infrastructure bills
6. **Mission-Driven:** Save lives, protect communities, enable climate adaptation

The Ask

Seeking: \$5M Seed to build founding team, develop MVP, and sign first pilots

Use of Funds: - 50% Engineering & Data Science - 25% Data Acquisition & Infrastructure
- 15% Sales & Partnerships - 10% Operations

Target Close: Q3 2026

Appendix

Comparable Company Analysis

Company	Last Raise	Valuation	Focus
Jupiter Intelligence	\$185M total	~\$500M	Climate analytics
Tomorrow.io	\$190M total	\$1B+	Weather prediction
Cape Analytics	\$84M total	~\$300M	Property imagery AI
Understory	\$35M total	~\$100M	Weather IoT

Key References

- NOAA National Weather Service Flood Data
- Munich Re NatCatSERVICE

- FEMA National Flood Insurance Program
- World Resources Institute Aqueduct
- Copernicus Climate Change Service

“The best time to build flood infrastructure was 20 years ago. The second best time is now — with AI.”

FloodMind AI — Because every flood should be predictable.

Generated by The Godfather / May 21, 2026